



THE UNIVERSITY
of EDINBURGH

Group 44 – EofE Project

B220071

B220808

B219095

B231358

Q1)

Variable	Obs	Mean	Std. Dev.	Min	Max
age	113180	41.061	7.308	30	55
black	113180	.085	.279	0	1
years schooling	113180	11.473	3.776	0	19
incwage	113180	1372.047	986.747	1	5001

Translating *higrade* to *years_schooling* quantifies the education level, making it easier to interpret in analyses. For example, a *higrade* of “12th grade” becomes 14 years of schooling¹. This continuous variable allows us to analyse changes in wages associated with each additional year of education.

The average age of participants is 41.061 years, indicating a middle-aged, working population. The standard deviation of 7.308 years shows a moderate age spread, helping us understand wage changes over a career. About 8.5% of participants are Black, suggesting most are White, which helps in analysing racial income disparities. Running regressions can reveal how race affects wages.

Participants have an average of 11.473 years of schooling, with a standard deviation of 3.776 years, indicating varied educational levels. Higher education generally correlates with higher productivity and wages. The average wage income is \$1372.047, with a high standard deviation of \$986.747, indicating significant wage variability influenced by age, education, and race. However, the data quantifies years of schooling up to 5 years of college but nothing after that. This could potentially limit the full accuracy of determining the exact years of education completed by the participant.

¹ In our code, we split years of schooling into separate categories to differentiate educational levels (see log file in appendix).

Q2)

log_wage	Coef.	St.Err.	t-value	p-value	[95% Conf	Interval]	Sig
years_schooling	.092	.001	143.16	0	.09	.093	***
Constant	5.862	.008	749.83	0	5.847	5.877	***
Mean dependent var		6.927	SD dependent var		0.879		
R-squared		0.153	Number of obs		113180		
F-test		20493.835	Prob > F		0.000		
Akaike crit. (AIC)		273129.339	Bayesian crit. (BIC)		273148.612		

*** $p < .01$, ** $p < .05$, * $p < .1$

The intercept of 5.862 is the wage level when the participant has not done any education. The coefficient of 0.092 indicates that each additional year of education is associated with a 9.2% wage increase, all things equal. The two coefficients have high t-values and p-values of 0, which indicates that years of education positively affect the wage levels of participants, showing that this relationship is statistically significant.

Q3)

Linear regression

log_wage	Coef.	St.Err.	t-value	p-value	[95% Conf	Interval]	Sig
years_schooling	.109	.001	139.73	0	.107	.11	***
potential_experience	.013	0	38.24	0	.012	.013	***
Constant	5.365	.015	354.03	0	5.335	5.394	***
Mean dependent var		6.927	SD dependent var		0.879		
R-squared		0.164	Number of obs		113180		
F-test		11110.503	Prob > F		0.000		
Akaike crit. (AIC)		271678.145	Bayesian crit. (BIC)		271707.055		

*** $p < .01$, ** $p < .05$, * $p < .1$

Using the Omitted Variable Bias (*OV/B*) formula, we see that the expected value of the estimated coefficient of *years_schooling* is " $\beta_1 + \beta_2 \hat{\partial}_1$ " (Wooldridge, 2020). The term $(\beta_2 \hat{\partial}_1)$ indicates an omitted variable present. For the estimate to be unbiased, $E(\beta'_1) = \beta_1$, which is not the case here. $\hat{\partial}_1$ shows the relationship between *potential_experience* and years of schooling.

Our result from using the *OV/B* formula shows that our regression in the previous question is biased as it omits experience, which is an important factor in determining wages. The bias is negative due to the inverse relationship between experience and years of schooling. More years spent in education suggests fewer years available for work experience.

The original model underestimated the effect of years of schooling on wages. The coefficient of *years_schooling* increased from 0.092 to 0.109, suggesting the effect of education on wages is stronger when accounting for experience. Adding potential experience to the original regression results in more reliable and accurate estimates of the coefficient of years of schooling. Therefore, adding the omitted variable is in line with our expectations.

Q4)

Age and years of schooling are correlated with each other as well as age and potential experience. Older people are likely to have more experience and education. This means we can't include these factors at the same time in our regression as the correlation between these factors can create a degree of multicollinearity, defined as when the predictor variables are highly correlated (Paul 2006). This is problematic for our model as multicollinearity can inflate the standard error and variance which leads to a reduction in our t-statistic. This can reduce the statistical significance of our findings and make our confidence intervals wider (Kim 2019).

Q5)

log_wage	Coef.	St.Err.	t-value	p-value	[95% Conf	Interval]	Sig
years_schooling	.109	.001	139.73	0	.107	.11	***
potential_experience	.017	.001	12.53	0	.015	.02	***
potential_experience ²	-.0000929	0	-3.39	.001	0	0	***
Constant	5.317	.021	257.20	0	5.276	5.357	***
Mean dependent var		6.927	SD dependent var			0.879	
R-squared		0.164	Number of obs			113180	
F-test		7411.517	Prob > F			0.000	
Akaike crit. (AIC)		271668.660	Bayesian crit. (BIC)			271707.207	

*** $p < .01$, ** $p < .05$, * $p < .1$

The effect of *potential_experience* on wages is signified by the coefficient of potential experience on log wage which is 0.017 which means that potential experience has a 1.7% effect increase on log wage. The coefficient of potential experience squared is -0.0000929 this shows that the return on potential experience to wages has a slight diminishing effect. The effect of the increase from 15 years to 16 years is 1.7%. The equation for determining the marginal increase from 15 to 16 years of experience is denoted by the: $[\Delta \log(\text{wage}) = \beta_1 + 2 \cdot \beta_2 \cdot \text{potential_experience}]$, which is calculated to be 0.014213 or a 1.4% increase on average. The level at which *potential_experience* is maximised where $\log(\text{wage})$ is derived and then equal to zero, which is denoted by the equation $[\text{experience} = -\beta_1 / 2 \cdot \beta_2]$. When calculated, this means that potential wages are roughly maximised at the age of 91.5.

Q6)

log_wage	Coef.	St.Err.	t-value	p-value	[95% Conf	Interval]	Sig
black	-.573	.009	-65.66	0	-.59	-.556	***
years_schooling	.096	.001	121.61	0	.094	.097	***
potential_experience	.017	.001	12.70	0	.014	.02	***
potential_experience~q	0	0	-4.60	0	0	0	***
Constant	5.536	.021	269.24	0	5.496	5.577	***
Mean dependent var		6.927	SD dependent var		0.879		
R-squared		0.195	Number of obs		113180		
F-test		6848.105	Prob > F		0.000		
Akaike crit. (AIC)		267439.448	Bayesian crit. (BIC)		267487.632		

*** $p < .01$, ** $p < .05$, * $p < .1$

The dummy variable, *black*, reflects the average difference in $\log(wage)$ between black and white respondents, holding other factors constant. The coefficient of -0.572 shows that black males earn 57.2% less than white males after controlling for potential experience and years of schooling. This could suggest there is a large degree of discrimination against black people in the labour force and shows that in our regression, race is the largest factor in determining wages. This is largely due to the social stratification of black Americans at the time wherein they were largely discriminated against in the workplace, with Jim Crow laws aiding these unfair practices with public rhetoric largely promoting this. Black people were portrayed as caricatures and less intelligent in media leading to further discrimination (Fremon 2014) (Poncza 2017).

Q7)



The above displays a regression line of log wages on schooling that is not explained by race or experience. Considering that the slope of this is equal to the initial coefficient of education in question 6, it can be inferred that the effect of education on wages excluding race and experience is the same. As suggested, the slope of the regression line of log wages on the residuals is equivalent to the slope coefficient for years of schooling in question 6, displaying a 9.6% increase in wages per additional year of schooling. This identical slope suggests that the economic return to education (how much wages increase with each additional year of schooling) remains consistent, even after controlling for the factors of race and experience. In practical terms, this means that the specific part of schooling that is not explained by race or experience still has a strong, positive impact on wages. Furthermore, Education independently contributes to higher earnings, and this effect is not significantly modified when it is isolated from the influence of other factors like race or years of experience. The above confirms the robustness of schooling's effect on wages, implying that the wage benefits of education are substantial and continue to exist even when accounting for demographic and experiential differences.

Q8)

Linear regression

log_wage	Coef.	St.Err.	t-value	p-value	[95% Conf	Interval]	Sig
black	-.755	.009	-86.54	0	-.772	-.738	***
graduated	.548	.006	89.55	0	.536	.56	***
potential_experience	.032	.001	22.34	0	.029	.035	***
potential_experience~q	-.001	0	-23.03	0	-.001	-.001	***
Constant	6.493	.018	368.12	0	6.458	6.527	***
Mean dependent var		6.927	SD dependent var			0.879	
R-squared		0.150	Number of obs			113180	
F-test		4989.302	Prob > F			0.000	
Akaike crit. (AIC)		273590.166	Bayesian crit. (BIC)			273638.349	

*** $p < .01$, ** $p < .05$, * $p < .1$

As displayed, the coefficient for the *graduated* dummy variable is 0.548. This shows that pupils who either graduated from high school or accomplished even higher education have on average, 54.8% higher wages than those who did not, *ceteris paribus*. The p-value of less than 0.01 suggests the statistical significance of high school education and any education beyond that on wages. In the previous question, years of schooling were a continuous variable to predict the effect of each additional year of schooling on log wages. The introduction of the dummy variable allows for a binary model of whether high school (or further education) has been achieved or not. This binary model allows one to identify a clearer general difference in the effect of education on wages. Policy analysis could also benefit from this binary approach. Significantly, introducing the dummy variable made the coefficient on *black* even more negative, suggesting that the dummy accounted for more of the effects of education on wages. Because this new model allows for better practical use and improves the statistical fit, the regression derived from this question is preferred to the regression in equation 6.

Q9)

Linear regression

log_wage	Coef.	St.Err.	t-value	p-value	[95% Conf	Interval]	Sig
black	-.481	.022	-22.13	0	-.524	-.439	***
years_schooling	.097	.001	118.52	0	.095	.098	***
potential_experience	.018	.001	12.96	0	.015	.02	***
potential_experience ~ q	0	0	-4.85	0	0	0	***
black_years_schooling ~ g	-.01	.002	-4.58	0	-.015	-.006	***
Constant	5.52	.021	264.54	0	5.479	5.561	***
Mean dependent var		6.927	SD dependent var		0.879		
R-squared		0.195	Number of obs		113180		
F-test		5483.636	Prob > F		0.000		
Akaike crit. (AIC)		267420.515	Bayesian crit. (BIC)		267478.335		

*** $p < .01$, ** $p < .05$, * $p < .1$

Generally, the regression results above indicated that the returns to education for white and black men are both positive. For white men, the coefficient for years of schooling indicates that each additional year is associated with a 9.7% increase in wages. Yet, black men only experience an increase of 8.7% in wages upon the completion of another year of schooling. This observed difference is emerging from the interaction term introduced, highlighting that for each additional year of education, the returns of education in terms of wages are 1% lower for black men than for white men (seen by *black_years_schooling*). When considering the p-values shown in the table, this result can be said to be statistically significant. A p-value of less than 0.01 suggests that the arising difference in returns to education for black and white men is unlikely to be by chance. Rather, the significance supports the idea that levels of education translate differently to wages for black and white men.

Q10)

For our preferred regression in Q8 to be unbiased and consistent, the first four Gauss-Markov assumptions (MLR.1 to MLR.4) must hold. With an additional assumption (MLR.5), the OLS estimator becomes the Best Linear Unbiased Estimator (BLUE).

The MLR.1 assumption of linearity in parameters holds in the preferred regression as it supports the use of a log-linear model for wage analysis, consistent with the Mincerian earnings equation (Mincer, 1974). This includes the quadratic term and dummy variable as they retain linearity in coefficients. The MLR.2 requires random sampling from the population. The 1940 U.S. Census data is a significant strength in ensuring consistency because it encompasses a wide range of employed individuals across different regions and demographics. However, the primary concern with the data is that it captures only those who were employed and reported wage income, thus resulting in an employment-based sample bias limiting generalisability. The preferred regression satisfies MLR.3 (no perfect multicollinearity among regressors) as there is no perfect linear relationship among the independent variables. This allows each variable to have a separate, identifiable effect on $\log(wage)$. The MLR.4 zero conditional mean assumption implies that no omitted variables are correlated with $\log(wage)$ and all the independent variables. The primary challenge here is the potential for *OV/B* such as ability, socioeconomic background, and quality of schooling. Griliches (1977) noted that *OV/B* can cause upward bias in estimating education returns when ability controls are missing. To improve the likelihood that the MLR.4 holds, we can use instrumental variables (IV), and proxy variables to enhance the reliability of the estimates. While MLR.5 (Homoskedasticity) may not fully hold due to real-world wage disparities influenced by race, education, and employment stability, this analysis operates under the assumption of homoskedasticity for simplicity; therefore, MLR.5 is considered to hold. In cases of heteroskedasticity, applying robust standard errors or using weighted least squares can correct for the unequal error variances, enhancing reliability.

Q11)

A linear probability model (LPM) is used for binary dependent variables, making the response probability linear in the parameters. Since the variable "nonwage" is binary, the homoskedasticity assumption is inherently violated. In an LPM, the variance of the error term is not constant as it depends on the predicted probability $p(1-p)$; as p varies, so does the variance, causing heteroskedasticity (Wooldridge, 2020). The White's and Breusch-Pagan/Cook-Weisberg tests can be used to detect heteroskedasticity in residuals. For the White's test, the null hypothesis (H_0) assumes homoskedasticity, while the alternative (H_a) indicates non-constant variance. The test result, with a chi-squared value of 1586.26 and a p-value of 0.0000, strongly rejects H_0 , indicating significant heteroskedasticity in the residuals.

To ensure reliable inference and correct estimation of standard errors, robust standard errors should be used to adjust for heteroskedasticity. Additionally, GLS estimation can be used, leading to WLS, which produces the Best Linear Unbiased Estimator. However, for binary models, MLE is more effective, as it directly models binary outcomes through logit or probit functions, producing probabilities bounded between 0 and 1.

Linear regression with nonwage as the binary outcome variable with robust standard errors.

nonwage	Coef.	St.Err.	t-value	p-value	[95% Conf	Interval]	Sig
black	-.035	.004	-9.62	0	-.042	-.028	***
years_schooling	.011	0	28.26	0	.01	.012	***
potential_experience	.005	.001	8.01	0	.004	.006	***
potential_experience ~ q	0	0	-0.93	.352	0	0	
Constant	-.074	.01	-7.37	0	-.093	-.054	***
Mean dependent var		0.167	SD dependent var			0.373	
R-squared		0.012	Number of obs			113180	
F-test		330.390	Prob > F			0.000	
Akaike crit. (AIC)		96752.272	Bayesian crit. (BIC)			96800.455	

*** $p < .01$, ** $p < .05$, * $p < .1$

The coefficients with and without robust standard errors are identical. Each additional year of schooling raises the probability of earning over \$50 in non-wage income by 1.1%, while being Black is associated with a 3.5% lower probability of earning more than \$50 in non-wage, holding other factors constant.

Bibliography

Fremon, D.K., 2014. *The Jim Crow laws and racism in United States history*. Enslow Publishing, LLC.

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Kim, J.H., 2019. Multicollinearity and misleading statistical results. *Korean journal of anesthesiology*, 72(6), pp.558-569.

Mincer, J. (1974) 'Chapter 5', in *Schooling, experience, and earnings*. New York: National Bureau of Economic Research, pp. 83–96.

Paul, R.K., 2006. Multicollinearity: Causes, effects and remedies. *LASRI, New Delhi*, 1(1), pp.58-65.

Poncza, E., 2017, *English Language and Literature*.

Wooldridge, J.M. (2020) 'Chapter 8 and 17', in *Introductory econometrics: A modern approach*. Boston, MA: Cengage, pp. 262–594.

Appendix

Q1)

```
-----  
-----  
-----  
      name: <unnamed>  
      log: /Users/mujtabasheikh/Desktop/EoE Stata/Group Project  
/Question 1.log  
      log type: text  
      opened on: 7 Nov 2024, 02:33:17  
  
. //Question 1  
/ is not a valid command name  
r(199);  
  
. generate age = 1940 - birthyr  
  
. keep if age >= 30 & age <= 55  
(89,581 observations deleted)  
  
.   
. generate black = (race == 2)  
  
. replace black = 0 if black == .  
(0 real changes made)  
  
.   
. generate years_schooling = higrade  
  
. replace years_schooling = 0 if higrade == 1 //No education  
1/ invalid name  
r(198);  
  
. replace years_schooling = 2 if inrange(higrade,2,3) //Pre-Elementary  
unknown function ()  
r(133);  
  
. replace years_schooling = 7 if inrange(higrade,4,8) //Elementary  
unknown function ()  
r(133);  
  
. replace years_schooling = 10 if inrange(higrade,9,11) //Middle School  
unknown function ()  
r(133);  
  
. replace years_schooling = 14 if inrange(higrade,12,15) //High School  
unknown function ()  
r(133);  
  
. replace years_schooling = 18 if inrange(higrade,16,19) //Undergrad  
unknown function ()  
r(133);  
  
. replace years_schooling = 19 if higrade == 20 //Postgrad
```

```
20/ invalid name
r(198);
```

```
.
. summarize age black years_schooling incwage
```

Variable	Obs	Mean	Std. dev.	Min	Max
age	113,180	41.06116	7.307931	30	55
black	113,180	.0847941	.2785763	0	1
years_scho~g	113,180	11.63072	3.758239	1	20
incwage	113,180	1372.047	986.7466	1	5001

```
.
. asdoc summarize age black years_schooling incwage,
save(Q1summary_stats.doc)
```

Variable	Obs	Mean	Std. dev.	Min	Max
age	113,180	41.06116	7.307931	30	55
black	113,180	.0847941	.2785763	0	1
years_scho~g	113,180	11.63072	3.758239	1	20
incwage	113,180	1372.047	986.7466	1	5001

Click to Open File: Q1summary_stats.doc

```
.
. log close
    name: <unnamed>
    log: /Users/mujtabasheikh/Desktop/EoE Stata/Group Project
/Question 1.log
    log type: text
    closed on: 7 Nov 2024, 02:33:24
```

```
-----
-----
-----
```

Q2)

```
-----
-----
name: <unnamed>
log: /Users/mujtabasheikh/Desktop/EoE Stata/Group Project
/Question 2.log
log type: text
opened on: 7 Nov 2024, 02:33:32

.
. generate log_wage = log(incwage)

. label var log_wage "Log(wage)"

.
. regress log_wage years_schooling

      Source |           SS          df           MS      Number of obs      =
113,180
-----+-----
20493.83
      Model |   13402.7386            1   13402.7386      Prob > F            =
0.0000
      Residual |   74017.1464       113,178   .653988818      R-squared            =
0.1533
-----+-----
0.1533
      Total |   87419.885       113,179   .772403759      Adj R-squared        =
.8087
      Root MSE

-----
-----
      log_wage | Coefficient   Std. err.      t    P>|t|    [95% conf.
interval]
-----+-----
years_schooling |   .091565    .0006396    143.16   0.000    .0903114
.0928186
      _cons |   5.862123    .0078179    749.83   0.000    5.8468
5.877446
-----
-----

.
. ssc install asdoc
checking asdoc consistency and verifying not already installed...
all files already exist and are up to date.

.
. ssc install outreg2
checking outreg2 consistency and verifying not already installed...
all files already exist and are up to date.

.
. asdoc regress log_wage years_schooling, save(Q2_Regression_results.doc)
```



```

Source |      SS      df      MS      Number of obs      =
113,180
-----+-----
20493.83
Model | 13402.7386      1 13402.7386 Prob > F      =
0.0000
Residual | 74017.1464 113,178 .653988818 R-squared      =
0.1533
-----+-----
0.1533
Total | 87419.885 113,179 .772403759 Root MSE      =
.8087

-----
log_wage | Coefficient Std. err.      t      P>|t|      [95% conf.
interval]
-----+-----
years_schooling | .091565 .0006396 143.16 0.000 .0903114
.0928186
_cons | 5.862123 .0078179 749.83 0.000 5.8468
5.877446
-----

(file Q2_Regression_results.doc not found)
Click to Open File: Q2_Regression_results.doc

.
. outreg2 using Q2Regression_results.doc, replace
Q2Regression_results.doc
dir : seeout

.
. log close
name: <unnamed>
log: /Users/mujtabasheikh/Desktop/EoE Stata/Group Project
/Question 2.log
log type: text
closed on: 7 Nov 2024, 02:33:37
-----
-----
-----

```

Q3)

```

-----
-----
-----
name: <unnamed>
log: /Users/mujtabasheikh/Desktop/EoE Stata/Group Project
/Question 3.log
log type: text
opened on: 7 Nov 2024, 02:33:42

.
. generate potential_experience = age - years_schooling - 6
. regress log_wage years_schooling potential_experience

      Source |           SS          df           MS       Number of obs   =
113,180
-----+-----
11110.50
      Model |  14347.0195            2       7173.50973   Prob > F             =
0.0000
      Residual |  73072.8656       113,177   .645651197   R-squared            =
0.1641
-----+-----
0.1641
      Total |   87419.885       113,179   .772403759   Root MSE            =
.80352

-----
-----
log_wage | Coefficient   Std. err.      t    P>|t|    [95%
conf. interval]
-----+-----
years_schooling |   .1087425   .0007782   139.73   0.000   .1072172
.1102678
potential_experience |   .0127085   .0003323    38.24   0.000   .0120572
.0133598
      _cons |    5.364571   .0151528   354.03   0.000   5.334871
5.39427
-----
-----

.
. asdoc regress log_wage years_schooling potential_experience,
save(Q3_Regression_results.doc)

      Source |           SS          df           MS       Number of obs   =
113,180
-----+-----
11110.50
      Model |  14347.0195            2       7173.50973   Prob > F             =
0.0000
      Residual |  73072.8656       113,177   .645651197   R-squared            =
0.1641

```

```

-----+-----
0.1641
Total |      87419.885      113,179      .772403759      Root MSE      =
.80352

-----
-----
log_wage | Coefficient   Std. err.      t    P>|t|      [95%
conf. interval]
-----+-----
years_schooling |      .1087425      .0007782    139.73    0.000      .1072172
.1102678
potential_experience |      .0127085      .0003323     38.24    0.000      .0120572
.0133598
_cons |      5.364571      .0151528    354.03    0.000      5.334871
5.39427

-----
(file Q3_Regression_results.doc not found)
Click to Open File:  Q3_Regression_results.doc

. outreg2 using Q3Regression_results.doc, replace
Q3Regression_results.doc
dir : seeout

.
. log close
      name:  <unnamed>
      log:   /Users/mujtabasheikh/Desktop/EoE Stata/Group Project
/Question 3.log
      log type:  text
      closed on:   7 Nov 2024, 02:33:43
-----
-----
-----

```

Q5)

```
-----
-----
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```

```
name: <unnamed>
log: /Users/mujtabasheikh/Desktop/EoE Stata/Group Project
/Question 5.log
log type: text
opened on: 7 Nov 2024, 02:33:55
```

```
.
. generate potential_experience_sq = potential_experience^2
```

```
.
. regress log_wage years_schooling potential_experience
potential_experience_sq
```

Source		SS	df	MS	Number of obs	=
113,180						
-----+-----					F(3, 113176)	=
7411.52						
Model		14354.4341	3	4784.81136	Prob > F	=
0.0000						
Residual		73065.451	113,176	.645591388	R-squared	=
0.1642						
-----+-----					Adj R-squared	=
0.1642						
Total		87419.885	113,179	.772403759	Root MSE	=
.80349						

```
-----
-----
log_wage | Coefficient Std. err. t P>|t| [95%
conf. interval]
-----+-----
years_schooling | .10874 .0007782 139.73 0.000
.1072148 .1102653
potential_experience | .0172288 .0013746 12.53 0.000
.0145346 .0199229
potential_experience_sq | -.0000929 .0000274 -3.39 0.001 -
.0001467 -.0000392
_cons | 5.316914 .020672 257.20 0.000
5.276397 5.357431
-----
-----
```

```
.
. asdoc regress log_wage years_schooling potential_experience
potential_experience_sq, replace save(Q5_Regression_results.doc)
```

Source		SS	df	MS	Number of obs	=
113,180						
-----+-----					F(3, 113176)	=
7411.52						

```

      Model | 14354.4341      3  4784.81136  Prob > F      =
0.0000
      Residual | 73065.451  113,176  .645591388  R-squared      =
0.1642
-----+-----
0.1642                               Adj R-squared      =
      Total | 87419.885  113,179  .772403759  Root MSE      =
.80349

```

```

-----
log_wage | Coefficient  Std. err.      t    P>|t|    [95%
conf. interval]
-----+-----
      years_schooling |      .10874   .0007782   139.73   0.000
.1072148   .1102653
      potential_experience |      .0172288   .0013746   12.53   0.000
.0145346   .0199229
potential_experience_sq |     -.0000929   .0000274    -3.39   0.001   -
.0001467   -.0000392
      _cons |      5.316914   .020672   257.20   0.000
5.276397   5.357431
-----

```

```

(file Q5_Regression_results.doc not found)
Click to Open File:  Q5_Regression_results.doc

```

```

. outreg2 using Q5Regression_results.doc, replace
Q5Regression_results.doc
dir : seeout

```

```

.
. log close
      name:  <unnamed>
      log:   /Users/mujtabasheikh/Desktop/EoE Stata/Group Project
/Question 5.log
      log type:  text
      closed on:  7 Nov 2024, 02:33:55

```

Q6)

```

-----
-----
-----
name: <unnamed>
log: /Users/mujtabasheikh/Desktop/EoE Stata/Group Project
/Question 6.log
log type: text
opened on: 7 Nov 2024, 02:34:07

```

```

.
. regress log_wage black years_schooling potential_experience
potential_experience_sq

```

Source	SS	df	MS	Number of obs	=
113,180					
-----+-----				F(4, 113175)	=
6848.11					
Model	17035.5428	4	4258.8857	Prob > F	=
0.0000					
Residual	70384.3422	113,175	.621907155	R-squared	=
0.1949					
-----+-----				Adj R-squared	=
0.1948					
Total	87419.885	113,179	.772403759	Root MSE	=
.78861					

	log_wage	Coefficient	Std. err.	t	P> t	[95% conf. interval]
-----+-----						
black	-.5726178	.0087211	-65.66	0.000	-	
.589711	-.5555246					
years_schooling	.0958849	.0007885	121.61	0.000		
.0943395	.0974303					
potential_experience	.0171397	.0013491	12.70	0.000		
.0144954	.019784					
potential_experience_sq	-.0001238	.0000269	-4.60	0.000	-	
.0001765	-.000071					
_cons	5.536386	.0205628	269.24	0.000		
5.496083	5.576689					

```

.
. asdoc regress log_wage black years_schooling potential_experience
potential_experience_sq, replace save(Q6_Regression_results.doc)

```

Source	SS	df	MS	Number of obs	=
113,180					
-----+-----				F(4, 113175)	=
6848.11					
Model	17035.5428	4	4258.8857	Prob > F	=
0.0000					
Residual	70384.3422	113,175	.621907155	R-squared	=
0.1949					

```

-----+-----
0.1948                               Adj R-squared   =
      Total |    87419.885    113,179    .772403759    Root MSE       =
.78861

-----
-----
      log_wage | Coefficient   Std. err.      t    P>|t|      [95%
conf. interval]
-----+-----
      black |   -.5726178   .0087211   -65.66   0.000      -
.589711   -.5555246
      years_schooling |   .0958849   .0007885   121.61   0.000
.0943395   .0974303
      potential_experience |   .0171397   .0013491    12.70   0.000
.0144954   .019784
      potential_experience_sq |  -.0001238   .0000269    -4.60   0.000      -
.0001765   -.000071
      _cons |    5.536386   .0205628   269.24   0.000
5.496083   5.576689
-----
-----
(file Q6_Regression_results.doc not found)
Click to Open File:  Q6_Regression_results.doc

.
. outreg2 using Q6Regression_results.doc, replace
Q6Regression_results.doc
dir : seeout

.
. log close
      name:  <unnamed>
      log:   /Users/mujtabasheikh/Desktop/EoE Stata/Group Project
/Question 6.log
      log type:  text
      closed on:   7 Nov 2024, 02:34:07
-----
-----
-----

```

Q7)

```
-----
-----
name: <unnamed>
log: /Users/mujtabasheikh/Desktop/EoE Stata/Group Project
/Question 7.log
log type: text
opened on: 7 Nov 2024, 02:34:47
```

```
.
. regress years_schooling black potential_experience
potential_experience_sq
```

Source	SS	df	MS	Number of obs	=
113,180					
-----+-----				F(3, 113176)	=
22561.73					
Model	598250.24	3	199416.747	Prob > F	=
0.0000					
Residual	1000330.74	113,176	8.83871796	R-squared	=
0.3742					
-----+-----				Adj R-squared	=
0.3742					
Total	1598580.98	113,179	14.1243604	Root MSE	=
2.973					

years_schooling	Coefficient	Std. err.	t	P> t	[95% conf. interval]
-----+-----					
black	-2.74644	.0318481	-86.24	0.000	-
2.808861 -2.684018					
potential_experience	-.230176	.0050399	-45.67	0.000	-
.2400542 -.2202978					
potential_experience_sq	-.0001789	.0001015	-1.76	0.078	-
.0003778 .00002					
_cons	17.36881	.0578258	300.36	0.000	
17.25547 17.48214					

```
.
. predict residuals
(option xb assumed; fitted values)

.
. twoway (scatter log_wage residuals) (lfit log_wage residuals)

.
. graph export "scatter_plot_log_wages_vs_residuals.pdf", replace
file /Users/mujtabasheikh/Desktop/EoE Stata/Group Project
/scatter_plot_log_wages_vs_residuals.pdf saved as PDF format

.
. log close
```



```
name: <unnamed>
log: /Users/mujtabasheikh/Desktop/EoE Stata/Group Project
/Question 7.log
log type: text
closed on: 7 Nov 2024, 02:34:57
```

```
-----
-----
-----
```

Q8)

```
-----
-----
-----
name: <unnamed>
log: /Users/mujtabasheikh/Desktop/EoE Stata/Group Project
/Question 8.log
log type: text
opened on: 7 Nov 2024, 02:35:27

.
. // Dropout dummy: 1 if years_schooling < 14 and 0 if otherwise
. generate dropout = (years_schooling < 14)

. label define label_dropout 1 "Dropped out before graduating from High
school" 0 "otherwise"

. label values dropout label_dropout

.
. //Graduated dummy: 1 if years_schoolin >= 14 and 0 if otherwise
. generate graduated = (years_schooling >= 14)

. label define label_graduated 1 "Graduated high school or even got
further education" 0 "otherwise"

. label values graduated label_graduated

.
. regress log_wage black dropout potential_experience
potential_experience_sq

Source | SS df MS Number of obs =
113,180
-----+-----
4989.30 F(4, 113175) =
Model | 13104.6936 4 3276.17341 Prob > F =
0.0000
Residual | 74315.1914 113,175 .656639641 R-squared =
0.1499
-----+-----
0.1499 Adj R-squared =
Total | 87419.885 113,179 .772403759 Root MSE =
.81033

-----
-----
log_wage | Coefficient Std. err. t P>|t| [95%
conf. interval]
-----+-----
black | -.7552867 .0087273 -86.54 0.000 -
.772392 -.7381814
```

```

              dropout |   -.5483031   .0061227   -89.55   0.000   -
.5603034   -.5363028
      potential_experience |   .0320469   .0014344    22.34   0.000
.0292354   .0348583
potential_experience_sq |   -.0006506   .0000282   -23.03   0.000   -
.0007059   -.0005952
              _cons |    7.041103   .0158631   443.87   0.000
7.010012    7.072194
-----
-----

```

```

.
. asdoc regress log_wage black dropout potential_experience
potential_experience_sq, replace save(Q8_Regression_results.doc)

```

```

      Source |           SS           df           MS       Number of obs   =
113,180
-----+-----
4989.30
      Model |   13104.6936             4   3276.17341       Prob > F           =
0.0000
      Residual |   74315.1914       113,175   .656639641       R-squared           =
0.1499
-----+-----
0.1499
      Total |   87419.885       113,179   .772403759       Root MSE           =
.81033

```

```

-----
-----
              log_wage | Coefficient   Std. err.      t    P>|t|     [95%
conf. interval]
-----+-----
              black |   -.7552867   .0087273   -86.54   0.000   -
.772392   -.7381814
              dropout |   -.5483031   .0061227   -89.55   0.000   -
.5603034   -.5363028
      potential_experience |   .0320469   .0014344    22.34   0.000
.0292354   .0348583
potential_experience_sq |   -.0006506   .0000282   -23.03   0.000   -
.0007059   -.0005952
              _cons |    7.041103   .0158631   443.87   0.000
7.010012    7.072194
-----
-----

```

```

(file Q8_Regression_results.doc not found)
Click to Open File:  Q8_Regression_results.doc

```

```

. outreg2 using Q8Regression_results.doc, replace
Q8Regression_results.doc
dir : seeout

```

```

.
. regress log_wage black graduated potential_experience
potential_experience_sq

```

Source		SS	df	MS	Number of obs	=	
113,180							
-----+-----						F(4, 113175)	=
4989.30							
Model		13104.6936	4	3276.17341	Prob > F	=	
0.0000							
Residual		74315.1914	113,175	.656639641	R-squared	=	
0.1499							
-----+-----						Adj R-squared	=
0.1499							
Total		87419.885	113,179	.772403759	Root MSE	=	
.81033							

log_wage			Coefficient	Std. err.	t	P> t	[95% conf. interval]
-----+-----							

	black		-.7552867	.0087273	-86.54	0.000	-
.772392	-.7381814						
	graduated		.5483031	.0061227	89.55	0.000	
.5363028	.5603034						
	potential_experience		.0320469	.0014344	22.34	0.000	
.0292354	.0348583						
potential_experience_sq		-.0006506	.0000282	-23.03	0.000	-	
.0007059	-.0005952						
	_cons		6.4928	.0176379	368.12	0.000	
6.45823	6.52737						

```
.
. asdoc regress log_wage black graduated potential_experience
potential_experience_sq, replace save(Q8b_Regression_results.doc)
```

Source		SS	df	MS	Number of obs	=	
113,180							
-----+-----						F(4, 113175)	=
4989.30							
Model		13104.6936	4	3276.17341	Prob > F	=	
0.0000							
Residual		74315.1914	113,175	.656639641	R-squared	=	
0.1499							
-----+-----						Adj R-squared	=
0.1499							
Total		87419.885	113,179	.772403759	Root MSE	=	
.81033							

log_wage			Coefficient	Std. err.	t	P> t	[95% conf. interval]
-----+-----							

	black		-.7552867	.0087273	-86.54	0.000	-
.772392	-.7381814						

	graduated	.5483031	.0061227	89.55	0.000	
.5363028	.5603034					
	potential_experience	.0320469	.0014344	22.34	0.000	
.0292354	.0348583					
potential_experience_sq		-.0006506	.0000282	-23.03	0.000	-
.0007059	-.0005952					
	_cons	6.4928	.0176379	368.12	0.000	
6.45823	6.52737					

(file Q8b_Regression_results.doc not found)

Click to Open File: Q8b_Regression_results.doc

. outreg2 using Q8bRegression_results.doc, replace
Q8bRegression_results.doc
dir : seeout

.

. log close

 name: <unnamed>

 log: /Users/mujtabasheikh/Desktop/EoE Stata/Group Project

/Question 8.log

 log type: text

 closed on: 7 Nov 2024, 02:35:28

Q9)

```

-----
-----
-----
name: <unnamed>
log: /Users/mujtabasheikh/Desktop/EoE Stata/Group Project
/Question 9.log
log type: text
opened on: 7 Nov 2024, 02:35:40

.
. capture drop black_years_schooling

. generate black_years_schooling = black*years_schooling

. label var black_years_schooling "black*years of schooling"

.
. regress log_wage black years_schooling potential_experience
potential_experience_sq black_years_schooling

Source |      SS          df           MS        Number of obs   =
113,180
-----+-----
5483.64
Model | 17048.5597          5      3409.71193    Prob > F           =
0.0000
Residual | 70371.3254    113,174      .621797633    R-squared           =
0.1950
-----+-----
0.1950
Adj R-squared           =
0.1950
Total | 87419.885    113,179      .772403759    Root MSE           =
.78854

-----
-----
log_wage | Coefficient  Std. err.      t    P>|t|     [95%
conf. interval]
-----+-----
black | -.4814304   .0217542   -22.13   0.000   -
.5240684   -.4387925
years_schooling | .0968705   .0008173   118.52   0.000
.0952685   .0984724
potential_experience | .0175204   .0013516   12.96   0.000
.0148713   .0201695
potential_experience_sq | -.0001308   .000027   -4.85   0.000   -
.0001836   -.0000779
black_years_schooling | -.0103315   .0022581   -4.58   0.000   -
.0147573   -.0059058
_cons | 5.520097   .0208669   264.54   0.000
5.479199   5.560996

```

```

.
. asdoc regress log_wage black years_schooling potential_experience
potential_experience_sq black_years_schooling, replace
save(Q9_Regression_results.doc)

```

Source	SS	df	MS	Number of obs	=
113,180					
				F(5, 113174)	=
5483.64					
Model	17048.5597	5	3409.71193	Prob > F	=
0.0000					
Residual	70371.3254	113,174	.621797633	R-squared	=
0.1950					
				Adj R-squared	=
0.1950					
Total	87419.885	113,179	.772403759	Root MSE	=
.78854					

	log_wage	Coefficient	Std. err.	t	P> t	[95% conf. interval]
black		-.4814304	.0217542	-22.13	0.000	-
.5240684		-.4387925				
years_schooling		.0968705	.0008173	118.52	0.000	
.0952685		.0984724				
potential_experience		.0175204	.0013516	12.96	0.000	
.0148713		.0201695				
potential_experience_sq		-.0001308	.000027	-4.85	0.000	-
.0001836		-.0000779				
black_years_schooling		-.0103315	.0022581	-4.58	0.000	-
.0147573		-.0059058				
_cons		5.520097	.0208669	264.54	0.000	
5.479199		5.560996				

```

(file Q9_Regression_results.doc not found)
Click to Open File: Q9_Regression_results.doc

```

```

. outreg2 using "tableQ9.xls", append label ctitle("log(wage)")
addstat(adj R^2, e(r2_a))
tableQ9.xls
dir : seeout

```

```

.
. test black_years_schooling

( 1)  black_years_schooling = 0

      F( 1,113174) =    20.93
      Prob > F =    0.0000

```

```
. log close
    name: <unnamed>
    log: /Users/mujtabasheikh/Desktop/EoE Stata/Group Project
/Question 9.log
    log type: text
closed on: 7 Nov 2024, 02:35:40
```

```
-----
-----
-----
```


Q11)

```

-----
-----
name: <unnamed>
log: /Users/mujtabasheikh/Desktop/EoE Stata/Group Project
/Question 11.log
log type: text
opened on: 7 Nov 2024, 02:36:11

```

```

.
. generate nonwage = (incnonwg == 2)

```

```

. replace nonwage = 0 if nonwage == .
(0 real changes made)

```

```

.
. regress nonwage black years_schooling potential_experience
potential_experience_sq

```

Source	SS	df	MS	Number of obs	=
113,180					
-----+-----				F(4, 113175)	=
343.38					
Model	189.060505	4	47.2651264	Prob > F	=
0.0000					
Residual	15578.1163	113,175	.137646268	R-squared	=
0.0120					
-----+-----				Adj R-squared	=
0.0120					
Total	15767.1768	113,179	.139311859	Root MSE	=
.37101					

nonwage	Coefficient	Std. err.	t	P> t	[95% conf. interval]
-----+-----					
black	-.0345901	.0041029	-8.43	0.000	-
.0426317 - .0265485					
years_schooling	.011142	.0003709	30.04	0.000	
.0104149 .011869					
potential_experience	.0052083	.0006347	8.21	0.000	
.0039642 .0064523					
potential_experience_sq	-.0000121	.0000127	-0.95	0.340	-
.0000369 .0000127					
_cons	-.0738159	.0096739	-7.63	0.000	-
.0927766 - .0548552					

```

.
.
. asdoc regress nonwage black years_schooling potential_experience
potential_experience_sq, replace save(Q11a_Regression_results.doc)

```

Source	SS	df	MS	Number of obs	=
113,180					
-----+-----					F(4, 113175)
343.38					
Model	189.060505	4	47.2651264	Prob > F	=
0.0000					
Residual	15578.1163	113,175	.137646268	R-squared	=
0.0120					
-----+-----					Adj R-squared
0.0120					
Total	15767.1768	113,179	.139311859	Root MSE	=
.37101					

	nonwage	Coefficient	Std. err.	t	P> t	[95% conf. interval]
-----+-----						
	black	-.0345901	.0041029	-8.43	0.000	-
.0426317	-.0265485					
	years_schooling	.011142	.0003709	30.04	0.000	
.0104149	.011869					
	potential_experience	.0052083	.0006347	8.21	0.000	
.0039642	.0064523					
	potential_experience_sq	-.0000121	.0000127	-0.95	0.340	-
.0000369	.0000127					
	_cons	-.0738159	.0096739	-7.63	0.000	-
.0927766	-.0548552					

(file Q11a_Regression_results.doc not found)
Click to Open File: Q11a_Regression_results.doc

. outreg2 using Q11aRegression_results.doc, replace
Q11aRegression_results.doc
dir : seeout

. estat imtest, white

White's test
H0: Homoskedasticity
Ha: Unrestricted heteroskedasticity

chi2(12) = 1586.26
Prob > chi2 = 0.0000

Cameron & Trivedi's decomposition of IM-test

Source	chi2	df	p
-----+-----			
Heteroskedasticity	1586.26	12	0.0000
Skewness	73217.77	4	0.0000
Kurtosis	35121.23	1	0.0000
-----+-----			

Total | 109925.26 17 0.0000

```
.
. regress nonwage black years_schooling potential_experience
potential_experience_sq, robust
```

Linear regression	Number of obs	=
113,180		
	F(4, 113175)	=
330.39		
	Prob > F	=
0.0000		
	R-squared	=
0.0120		
	Root MSE	=
.37101		

	nonwage	Coefficient	Robust std. err.	t	P> t	[95% conf. interval]
black		-.0345901	.0035949	-9.62	0.000	-
years_schooling		.011142	.0003942	28.26	0.000	
potential_experience		.0052083	.0006504	8.01	0.000	
potential_experience_sq		-.0000121	.000013	-0.93	0.352	-
_cons		-.0738159	.010018	-7.37	0.000	-

```
.
. asdoc regress nonwage black years_schooling potential_experience
potential_experience_sq, robust replace save(Q11b_Regression_results.doc)
```

Linear regression	Number of obs	=
113,180		
	F(4, 113175)	=
330.39		
	Prob > F	=
0.0000		
	R-squared	=
0.0120		
	Root MSE	=
.37101		

	nonwage	Coefficient	Robust std. err.	t	P> t	[95% conf. interval]
--	---------	-------------	---------------------	---	------	-------------------------

```

-----+-----
-----
      black |  -.0345901   .0035949   -9.62   0.000   -
.0416359   -.0275442
      years_schooling |   .011142   .0003942   28.26   0.000
.0103693   .0119146
      potential_experience |   .0052083   .0006504    8.01   0.000
.0039336   .006483
potential_experience_sq |  -.0000121   .000013   -0.93   0.352   -
.0000375   .0000133
              _cons |  -.0738159   .010018   -7.37   0.000   -
.0934511   -.0541807
-----
-----
(file Q11b_Regression_results.doc not found)
Click to Open File:  Q11b_Regression_results.doc

.
. outreg2 using Q11bRegression_results.doc, replace
Q11bRegression_results.doc
dir : seeout

.
. log close
      name:  <unnamed>
      log:   /Users/mujtabasheikh/Desktop/EoE Stata/Group Project
/Question 11.log
      log type:  text
      closed on:   7 Nov 2024, 02:36:11
-----
-----
-----

```